

PHASE 3 – PROJECT DESIGN

3.1 Project Architecture

AI BlogNest follows a modular backend architecture where incoming API requests pass through different processing layers before reaching the database or AI services.

The major architectural components are:

API Interface

The API interface provides endpoints through which clients can communicate with the backend. Tools such as Postman or Thunder Client can be used to send API requests and inspect responses.

Express Server

The Express server manages incoming HTTP requests and provides routing functionality for GET, POST, PUT, and DELETE operations.

Authentication Middleware

Authentication middleware verifies the identity of users using JWT tokens.

Authorization Layer

The authorization layer verifies whether the authenticated user has the required role to access a protected resource.

Controller Layer

Controllers process incoming requests, execute the required business logic, communicate with models, and return appropriate responses.

Database Layer

Mongoose provides the connection between the Node.js application and MongoDB. It handles schemas, validation, and database operations.

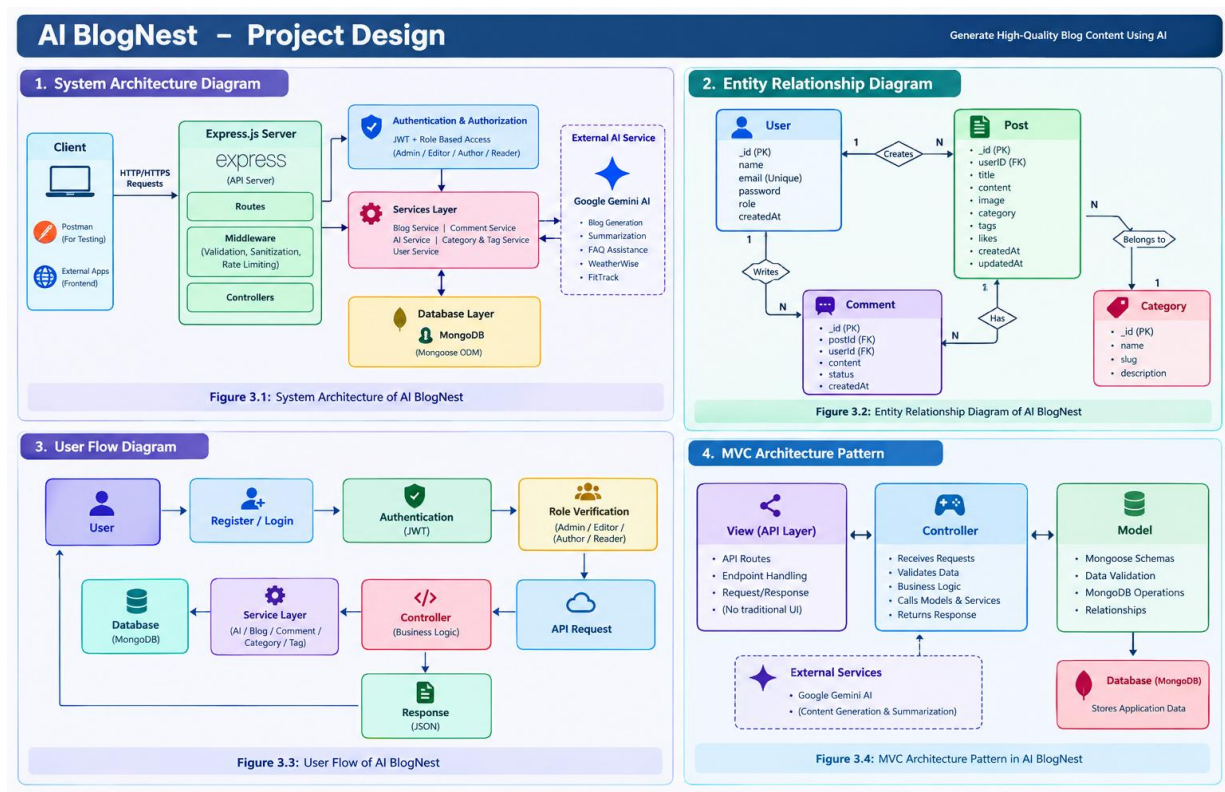
AI Integration Layer

The AI integration layer communicates with Google Gemini AI for intelligent content generation and summarization.

3.2 Main Modules

The major modules of AI BlogNest are:

1. User Authentication Module
2. User and Role Management Module
3. Blog Management Module
4. Category and Tag Module
5. Comment Management Module
6. AI Blog Generation Module
7. AI Summarization Module
8. Search Module
9. Media Metadata Module
10. Error Handling and Security Module



3.3 Entity Relationship Design

User Entity

The User entity stores information related to registered users.

Main attributes include:

- `_id`
- `name` `email`
- `password`
- `role`

Post Entity

The Post entity contains information about blog articles.

Main attributes include:

- `_id`
- `userID`
- `name`
- `photo`
- `message` `likes`
- `timestamps`

Relationships

User → Post

One user can create multiple blog posts.

User → Comment

A user can create multiple comments.

Post → Comment

A single blog post can contain multiple comments.

3.4 Role Design

Administrator

The administrator manages users, roles, comments, and other protected system operations.

Editor

The editor manages editorial workflows, reviews content, and handles publishing-related activities.

Author

The author creates and manages blog content associated with their account.

Reader

The reader accesses published blog content and participates through comments.

3.5 MVC Architecture

The system follows an MVC-oriented structure.

Model

The Model layer defines the data structure using Mongoose schemas and communicates with MongoDB.

Controller

The Controller layer receives requests from routes, processes business logic, interacts with models, and generates responses.

View / API Layer

Since AI BlogNest is primarily a backend REST API, the traditional frontend View layer is replaced by API routes through which external clients communicate with the application.